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Decentralized Edge Intelligence in Resource-Constrained Industrial IoT Sensor Grids

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ABSTRACT

Industrial Internet of Things (IIoT) deployments require sub-second anomaly detection while operating across low-bandwidth wireless mesh channels. We present a decentralized edge intelligence framework that distributes lightweight neural telemetry classifiers directly across sensor microcontrollers. Sensors execute local temporal inference and only transmit compressed consensus embeddings when anomalous vibration or thermal signatures are detected. Field trials across a 120-node factory grid demonstrate an 84% reduction in uplink bandwidth consumption and a sub-20ms anomaly response time. However, local node failures in cluster-head elections can temporarily cause telemetry blind spots during network topology reconfigurations.

Keywords— Internet of Things, Edge Computing, Industrial IoT, Sensor Networks

INTRODUCTION

Modern manufacturing plants generate gigabytes of sensor telemetry per second. Streaming all raw data to cloud dashboards causes severe bandwidth saturation and latency penalties that preclude real-time safety shutoffs.

METHODOLOGY

We embed 8-bit quantized temporal convolutional networks onto ARM Cortex edge motes. Nodes form localized clusters using dynamic gossip protocols, aggregating intermediate anomaly confidence scores before alerting supervisors.

RESULTS

In an operating manufacturing plant with 120 sensor nodes, the system operated continuously for six months, reducing gateway uplink traffic from 14.2 GB/day to 2.1 GB/day while detecting 100% of bearing failure anomalies.

LIMITATIONS

Dynamic mesh reorganization during node battery depletion introduces transient routing loops, momentarily delaying alarm propagation by up to 450 milliseconds.

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